

Before the introduction of the metric system, one may divide the history of Indian systems of measurement into three main periods: the pre-Akbar period, the period of the Akbar system, and the British colonial period.

During the Indian pre ancient period, weights and measure systems varied from region to region, commodity to commodity, and rural to urban areas. The weights were based on the weight of various seeds (in particular the wheat berry and Ratti) and lengths were based on the length of arms and width of fingers. During his reign, the Mughal emperor Akbar realized a need for a uniform system, and used the weight of the barley corn as a standard. This did not replace the existing system; rather, it simply added another system of measurement.

When the British first began trading in India, they accepted barley corn as a unit for weighing gold. Eventually, the British introduced their own system for weighing gold. In 1956, the government of independent India passed the Standards of Weights Act, which would come into effect in 1958. The metric system was made mandatory for weights in October 1960, and for measures in April 1962.

Conversion

In 1956, for metrication, the Indian government defined the Standards of Measurements Act (No. 89 of 1956, amended 1960, 1964) as follows:

The current definitions as per the UN are:

Ancient system

These are the weights and measures popular in North India before the adoption of the metric system. There were different systems in Bengal, the Presidency of Madras, and Bombay. The following nomenclature was prevalent in North India until the metric system was established:

4 Chāwal (grain of rice) = 1 Dhan (weight of one wheat berry)

4 Dhan = 1 Ratti (seed of the 'Abrus precatorius' plant, called 'Surkh' by Abul Fazl in Ain-i-Akbari.).

8 Ratti = 1 Masha

3 Masha (24 Ratti/96 Dhan) = 1 Tak

4 Tak (12 Masha/96 Ratti) = 1 Bhari

Conversion

1 Bhari = 11.66375 gram

3.75 Troy ounce = 10 Bhari

Weight of 64 Dhan (Wheat berries) = Weight of 45 Jau (Barley corns)

Weight of 1 Barley corn = 64.79891 milligrams

Commodity weight system

1 Bhari = 4 Siki

1 Kancha = 5 Siki

1 Chhataank = 4 Kancha

1 Chhataank = 5 Bhari

1 Adh-pav = 2 Chhatank = 1/8 Seer

1 Pav = 2 Adh-pav = 1/4 Seer (Pav means 1/4)

The unit pav is still used to this date however, it has been modified to "a fourth of a kilogram".

1 Adher = 2 Pav = 1/2 Seer

In Hindi 1/2 Seer = Adha (1/2) Seer, or Adher

1 Ser = 2 Adher = 4 Pav = 16 Chattank = 80 Tola = 933.1 grams

1 Savaser = 1 Ser + 1 Pav (1 1/4 Seer)

1 Savaser weighed 100 Imperial rupees

In Hindi $1\frac{1}{4}$ Seer = Sava ($1\frac{1}{4}$) Seer, or Savaser

1 Dhaser = 2 Savaser = $2\frac{1}{2}$ Seer

In Hindi $2\frac{1}{2}$ Seer = Dhai ($2\frac{1}{2}$) Seer, or Dhaser

1 Paseri = 2 Adisari = 5 Seer

In Hindi 5 Seer = Panch (5) Seer, or Paseri for short

1 Daseri = 2 Pasri = 10 Seer

In Hindi 10 Seer = Das (10) Seer, or Daseri for short

1 Maund (maan or man[माण]) = 4 Daseri = 8 Pasri = 40 Seer

Rice and grains volume measures

Grains were not weighed. Special hour-glass shaped measure were used to determine the volume.

Smallest unit = 1 Nilve

2 Nilve = 1 Kolve

2 Kolve = 1 Chipte (about quarter litre)

2 Chipte = 1 Mapte (about half litre)

2 Mapte = 1 Ser (about one litre)

Liquid volume measures

These were hour glass shaped measure used for Milk, Ghee, Oils. The bottom was round like an inverted dome, the top was like flared rim. This shape helped in pouring the liquids.

4 Chhataank = 1 Pav

4 Pav = 1 Seer

40 Seer = 1 Maund

Length measure

Measure of length is Gaz. To interpret Gaz, depends on what one is measuring and where they are. Bengal: 36", Bombay: 27", Madras: 33", Government Average: 33". The hand measurements were used.

Anguli (width of 3 fingers) = 1 Girah

8 Girah = 1 Hath (elbow to the end of the middle finger, approximately 18")

5 5/6 Hath = One Kathi

20 Kathi = One Pand

1 Pand= 1 Beesa

20 Pand = One Begah

2 Hath = 1 Gaz

3 Gaz = Two Karam

3 Karams = 1 Kan

3 Square Kans = 1 Marla

20 Marlas = 1 Kanal

8 Kanals = 1 Ghamaon

9 Kanals 12 Marlas = 1 Acre

4 Kanals = 1 Begah

Medieval system

Akbar weights and measures

Akbar standardised weights and measurements using a barley corn (Jau). For weights, he used the weight of a Jau, while the width of a Jau set the standard for length.

1. Length: Ilahi Gaz (33 to 34 inches or 840 to 860 millimetres); 1 Gaz = 16 Grehs; 1 Greh = 2

pais

At the time of Shah Jahan there existed three different Gaz:

Shahi gaz = 101.6 cm

Shahijahani or Lashkari = 95.85 cm

Aleppo gaz = 67.73 cm

Commodity weight: Ser = 637.74 grams

Commodity Spices: The Dam was a copper coin used as a weight as well as currency. 1 Dam = 20 grams

Gold and Expensive Spices: Misqal = 6.22 grams

Weights before 1833

8 rattīs = 1 māshā (= 0.907(2) gram)

12 māshās = 1 tolā (= 10.886227 gram)

80 tolas = 1 ser (= 870.89816 gram)

40 sers = 1 maund (= 34 kg 8 hg 3 dag 5g 9 dg 2.6 cg /34.835926 kilograms)

1 rattī = 1.75 grains (= 0.11339825 gram/113 milligrams 398 1/4 micrograms 4 attograms) (1 grain = 0.064799 gram)

From 1833 the rupee and tolā weight was fixed at 180 grains, i.e. 11.66382 grams. Hence the weight of 1 maund increased to 37.324224 kilogram. Traditionally one maund represented the weight unit for goods which could be carried over some distance by porters or pack animals.

British system

References

1 coss= 2000 yards

Notes

Sources

Prinsep, James (editor Edward Thomas): *Essays on Indian Antiquities, Historic, Numismatic, and Palaeographic, of the late James Prinsep, F.R.S., to which are added his Useful Tables illustrative of Indian History, Chronology, Modern Coinages, Weights, Measures etc.* Two Volumes, Reprint, Indological Book House, Delhi and Varanasi, 1971. Originally published in London, 1858.